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B.Tech. Degree VII Semester Examination November 2017

CS 1702 ARTIFICIAL INTELLIGENCE (2012 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) What are software agents?
 (b) Write short note on agent topology.
 (c) Explain the mini max procedure used in game playing.
 (d) Explain the different 'heuristic search' techniques.
 (e) Define the term clause. List an example for it.
 (f) Explain the unification algorithm.
 (g) Explain the knowledge representation mechanism of scripts.
 (h) Define Transformational Grammar.

PART B

(4 × 15 = 60)

- II. Write short notes on:
 (a) Agent characteristics. (b) Agent oriented programming. (7 + 8 = 15)

OR

- III. Explain how intelligent agents are implemented in Java. (15)

- IV. (a) Explain the steepest-ascent hill climbing algorithm. List down the cases in which it may end up failing to find goal state. How can those problems be circumvented? (10)
 (b) Explain 'Generate-and-Test' algorithm. (5)

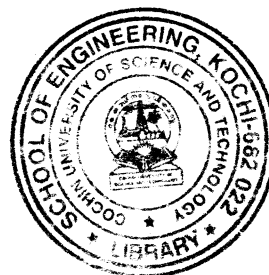
OR

- V. (a) For the following cryptarithmic puzzle, what will be the algorithm you can adopt to reach at a solution? Explain the algorithm with respect to the puzzle: (10)

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- (b) Define a production system. (5)



(P.T.O.)

- VI. List the steps for converting an English statement represented in wff to clause form. Explain the steps for converting the following statement to clause form: "All Romans who know Marcus either hate Caesar or think that anyone who hates anyone is crazy". (15)

OR

- VII. (a) Consider the following sentences: (10)
- (i) Anyone whom Mary loves is a football star.
 - (ii) Any student who does not pass does not play.
 - (iii) John is a student.
 - (iv) Any student who does not study does not pass.
 - (v) Anyone who does not play is not a football star.
- Prove that "If John does not study, then Mary does not love John" using Resolution-Refutation.
- (b) What are Skolem constants and Skolem functions? (5)

- VIII. (a) Explain the various steps in Natural language processing. (7)
- (b) Distinguish between the knowledge representation mechanism of frames and semantic net. (8)

OR

- IX. (a) What is meant by learning? Detail any two methods of learning. (7)
- (b) Write short notes on: (4+4=8)
- (i) Case grammar.
 - (ii) Fuzzy Logic.
